Aaron Legg
all288@cornell.edu | https://github.com/aaronlegg1| linkedin.com/in/aaronleelegg | cell: 602.772.0270 | Scottsdale, AZ

EDUCATION

Cornell University, College of Engineering **Aug 2024-May 2028**
BS Chemical Engineering and Operations Research **Ithaca, NY**

Scottsdale Community College **Aug 2023-May 2024**
Associates of Science in General Engineering, Mathematics, General Studies **Scottsdale, AZ**
Related Coursework: Probability and Statistics I & II • Data Structures • Stochastic Calculus/Processes • Foundations of AI Decisions

RELEVANT EXPERIENCE

Fry's Food Stores **June 2026-Aug 2026**
Operations Intern – Supply Chain Management **Scottsdale, AZ**
- Optimized ordering efficiency by 18% by implementing an AI agent to autonomously order products that were out of stock
- Reduced void transactions per shift by 40% by training and coaching 20+ associates on tender, coupon, and price-check standards
- Drove till variance to under $2 per shift by auditing 5+ tills daily by enforcing scanning errors and procedures across 3 registers

Tiny Smiles **May 2025-Aug 2025**
Risk Management Intern **Scottsdale, AZ**
- Improved portfolio returns by 7.13% by performing Monte Carlo risk simulations and analytics on six-figure investment accounts
- Reduced downside risk exposure by 18% by executing performance attribution analysis and asset allocation optimization in Excel
- Strengthened governance quality by delivering bi-weekly P&L summaries and investment strategy reviews to the executive board

Cornell Trading **Aug 2026-Current**
Quantitative Chair – NME Instructor **Ithaca, NY**
- Trained 43 new members by designing and instructing an 8-week quant curriculum spanning Python, stats-arbitrage, back testing
- Directed 6 research teams, standardized a shared Python back-testing framework that cut strategy-development time by 30%
- Mentored 43 members through a paper trading competition, with top teams reaching a 1.5 Sharpe ratio over a 10-week period

Systems Engineering **Aug 2026-Current**
Teaching Assistant – SYSTN 6170 Quantitative Foundations of Systems Engineering **Ithaca, NY**
- Led weekly office hours and recitations for 30 systems engineering PhD students, resolving 100+ questions on linear optimization
- Graded 5 problem sets and 5 modeling case studies for 30 students, focusing on optimization formulas in Python and MATLAB
- Mentored 8 student teams through 25-page final projects, advising on model formulation and validation of policy interventions

ACADEMIC PROJECTS

HMM Regime Trading Bot **Jan 2025**
- Eliminated look-ahead bias by implementing forward algorithm filtered inference in a Gaussian HMM with BIC model selection
- Capped max drawdown under 10% via volatility rank allocation (60-95% exposure, 1.25x leverage) across 10 equities
- Built a separate risk layer with tiered circuit breakers, 60-day correlation caps, and 100-run Monte Carlo cash-gap stress testing

Monte Carlo Option Pricer **Jan 2025**
- Achieved 0.01% pricing error by developing a high-performance derivatives pricing engine simulating 1M+ stochastic paths
- Improved computation speed by 35% by optimizing random sampling algorithms and decreasing time/space complexity to O(N)
- Validated model reliability by implementing put-call parity testing and by comparing outputs to the weekly average EFX futures

Voice Controlled Ledger **Mar 2026**
- Achieved sub-3-second voice-to-transaction latency by architecting a 3-model AI pipeline (Whisper, GPT, Claude) on Fast API
- Reduced manual ledger entry time by 80% by building a natural language interface parsing amount and FX across 99 languages
- Validated full-stack reliability by integrating 5 Google APIs (OAuth 2.0, sheets, docs, calendar) via a unified token refresh helper

Cornell Autonomous Boat Project Team **Jan 2025-Present**
- Raised $12,500 in sponsorship capital by executing targeted alum outreach, cold-call campaigns, and a multi-tier funding pipeline
- Improved capital efficiency by 15% by applying budget forecasting, ROI optimization modeling, and variance analysis
- Expanded alumni engagement by delivering performance reporting newsletters and sponsor-tier updates to 300+ former members

SKILLS & INTERESTS

Programming Languages: Python, C/C++, R, MATLAB, Java, JavaScript, MS Excel, MS Word, SQL, GIT, Pandas, NumPy, Julia
Interests: Poker, Finance, Strategy Games (Chess, Clash Royale), Pickleball, Violin, Football, Zetamac, snowboarding, Mandarin